ONYX Archiver

Secure Data Exchange and Cryptographic Archiving System

Description

ONYX Archiver is a desktop application designed for creating secure archives with end-to-end encryption support.
Only an authorized recipient who has completed the key exchange process can access the data.

The project addresses the problem of secure file transfer without relying on third-party services or infrastructure.

Key Features

- **Custom Format (.onx)**
  Hybrid encryption of files and folders with protection against interception and traffic analysis
- **Peer-to-Peer Handshake (.onxk)**
  Public key exchange without servers, enabling true end-to-end encryption
- **Secure User Profile:**
  Implementation based on:
  - Local database (SQLite)
  - Argon2id for password protection
  - X25519 for key exchange
  - XChaCha20-Poly1305 for encryption
- **Contacts (Peers) System**
  Trusted recipient management:
  - add
  - verify
  - remove

Security Features

- **Zero Trust Architecture**
  Data is encrypted on the sender's side and never transmitted in plaintext
- **Hybrid Cryptography**
  - X25519 (ECDH)
  - HKDF (key derivation)
  - XChaCha20-Poly1305 (encryption)
  - Ed25519 (digital signatures)

- **Master Key Protection**
  - Derivation using Argon2id
  - Encryption before storage
  - Use of Windows DPAPI during active session
  - Secure cleanup on logout
- **Metadata Protection**
  Dynamic Key ID based on SHA-512 prevents correlation and tracking of user interactions

Architecture

The project is built using modern development approaches:
- **MVVM** — separation of UI and business logic
- **Dependency Injection** — flexibility and testability
- **Asynchronous processing** — UI remains responsive during file and encryption operations
- **Modular design** — easy extensibility of algorithms and features

Update System

ONYX Archiver includes a full-featured secure update infrastructure consisting of multiple components:

1. UpdateServer
- Central server providing application updates
- Stores versions, metadata, and digital signatures

2. ONYX Archiver
- Checks for new versions on UpdateServer
- If a newer version is available, prompts the user for download
- Downloads the update package and verifies its integrity using a digital signature (UpdateSigner)

3. UpdateAgent
- Separate application responsible for installing updates
- After receiving the update package, it performs:
  - backup of the current version
  - safe file replacement
  - restarting ONYX Archiver

4. UpdateSigner

- Console tool for generating digital signatures for updates
- Produces update metadata:
  - file hash
  - digital signature
  - public key
  - archive size
- Ensures integrity verification and protection against tampering

Implementation highlights:

- Clear separation of responsibilities between client, installer, and server
- Full integrity and signature verification
- Minimizes the risk of application corruption during updates

Technologies

- C# / .NET 8
- WPF
- Entity Framework Core
- SQLite
- ZStandard (compression)
- Cryptography (NSec + Bouncy Castle):
  - X25519
  - XChaCha20-Poly1305
  - Ed25519
  - Argon2id
  - HKDF

Implementation Highlights

- Asynchronous processing of large files
- Real-time progress reporting
- Flexible key and session management system
- Extensible architecture

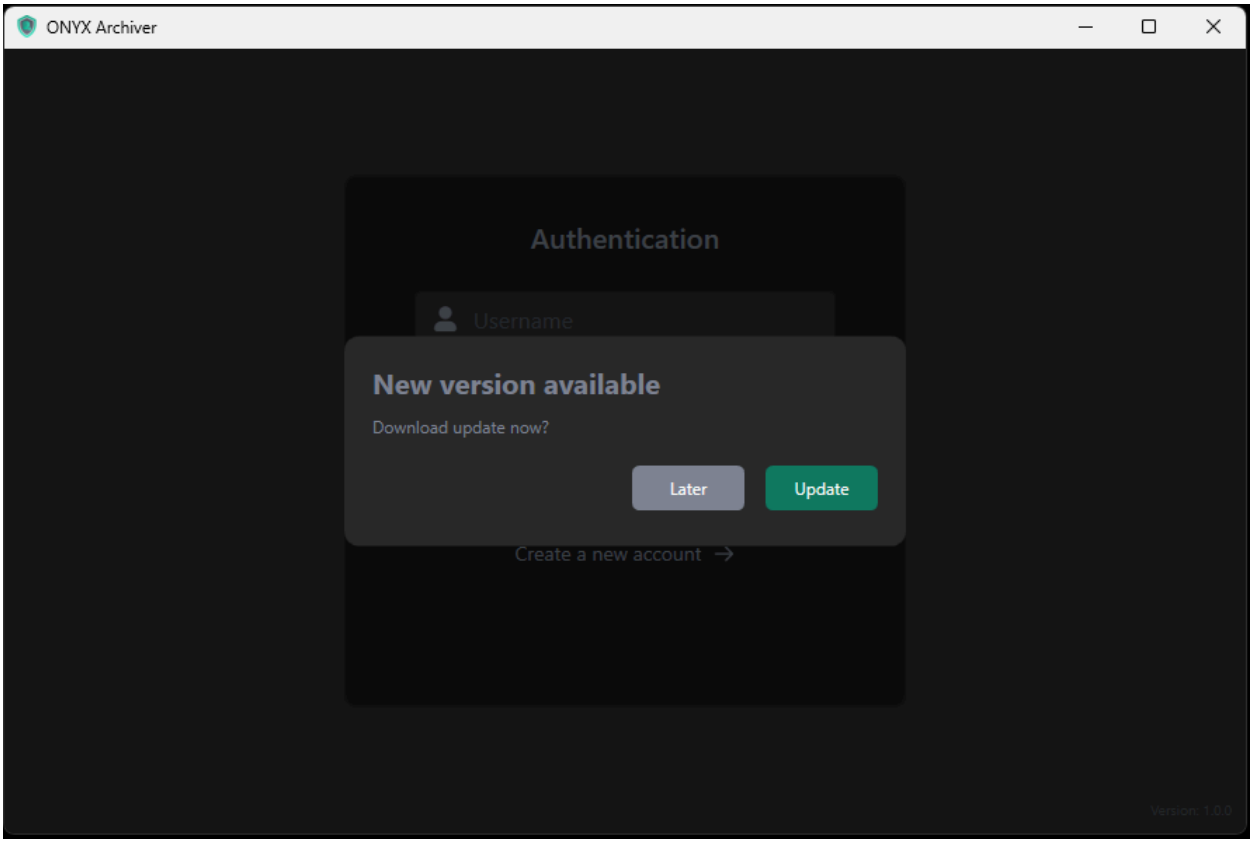
Summary

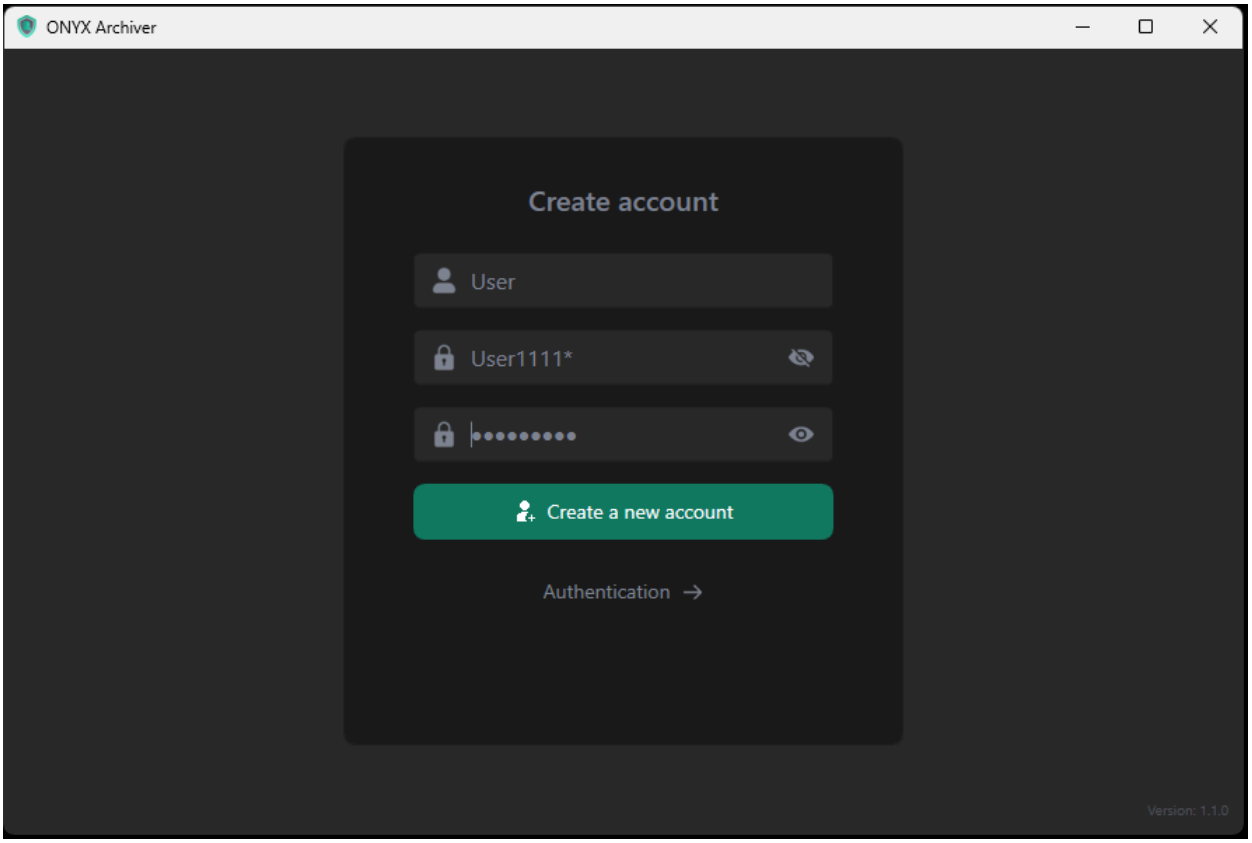
ONYX Archiver is not just an archiver, but a complete secure data exchange system demonstrating the practical application of modern cryptographic techniques and architectural patterns in desktop development.

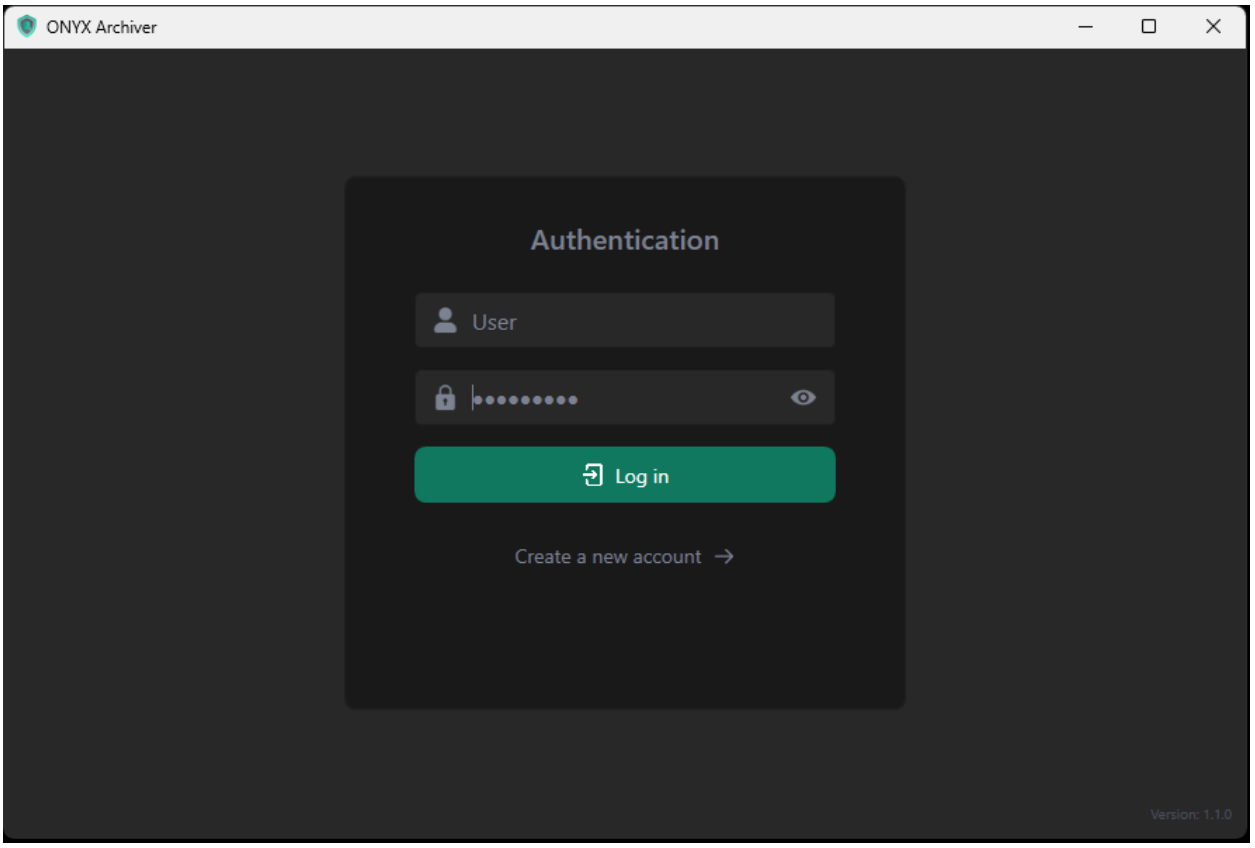
Getting Started

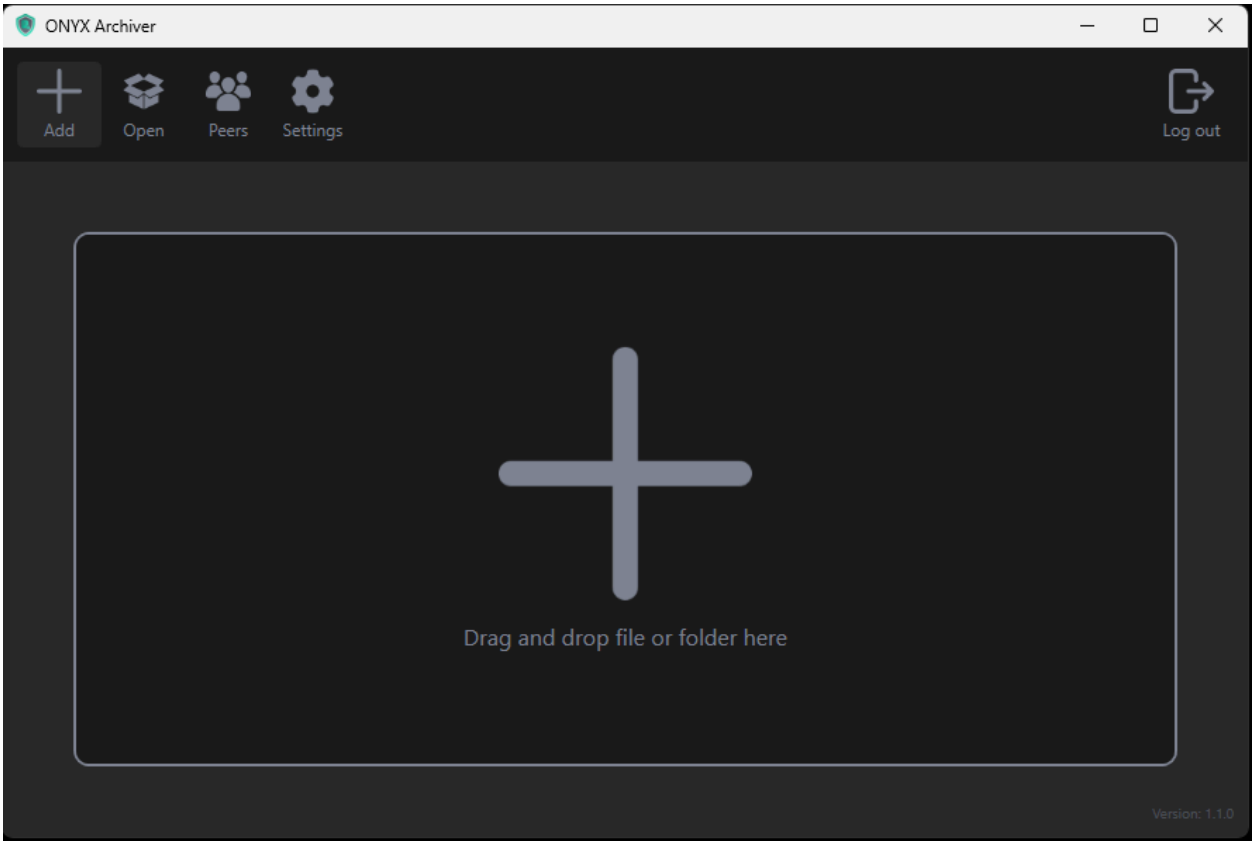
1. Download the latest version from Releases
2. Launch the application
3. Create a profile and perform key exchange

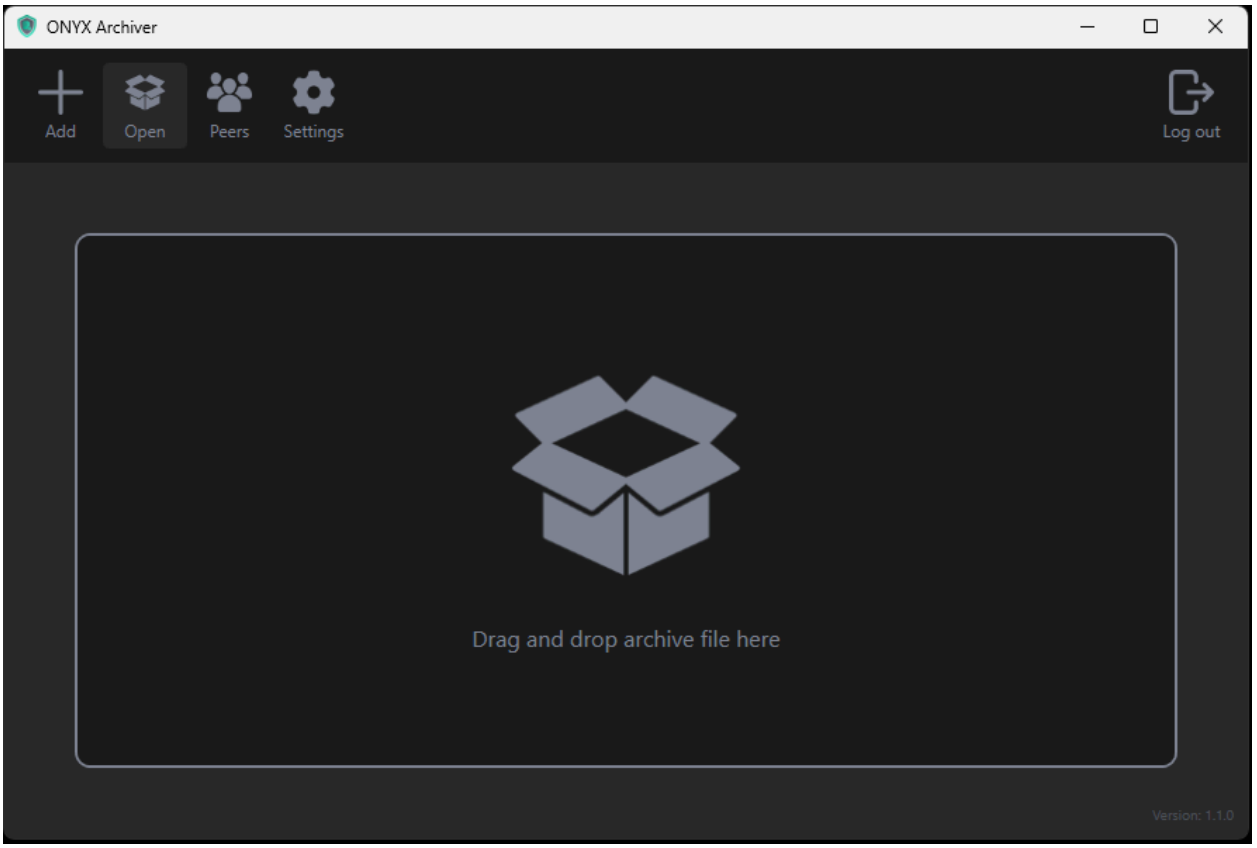
Screenshots

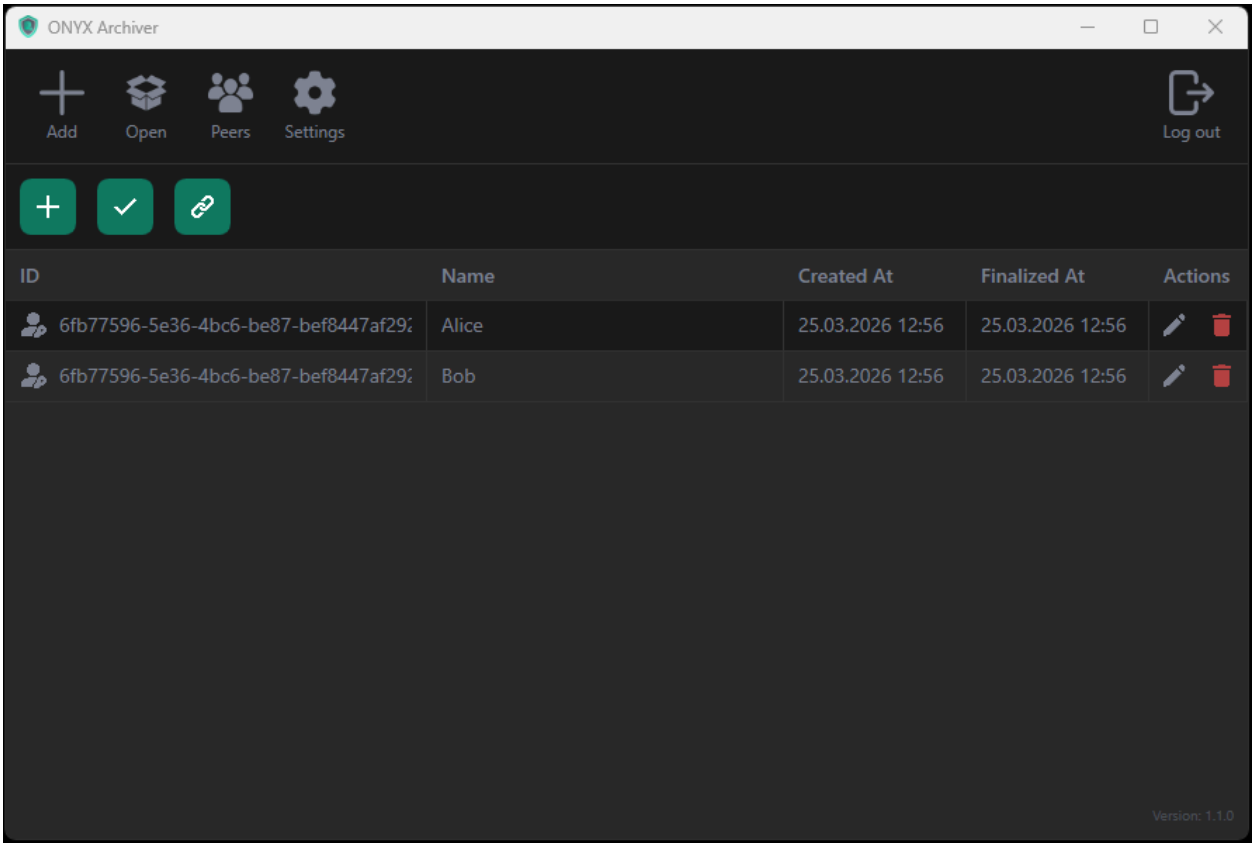


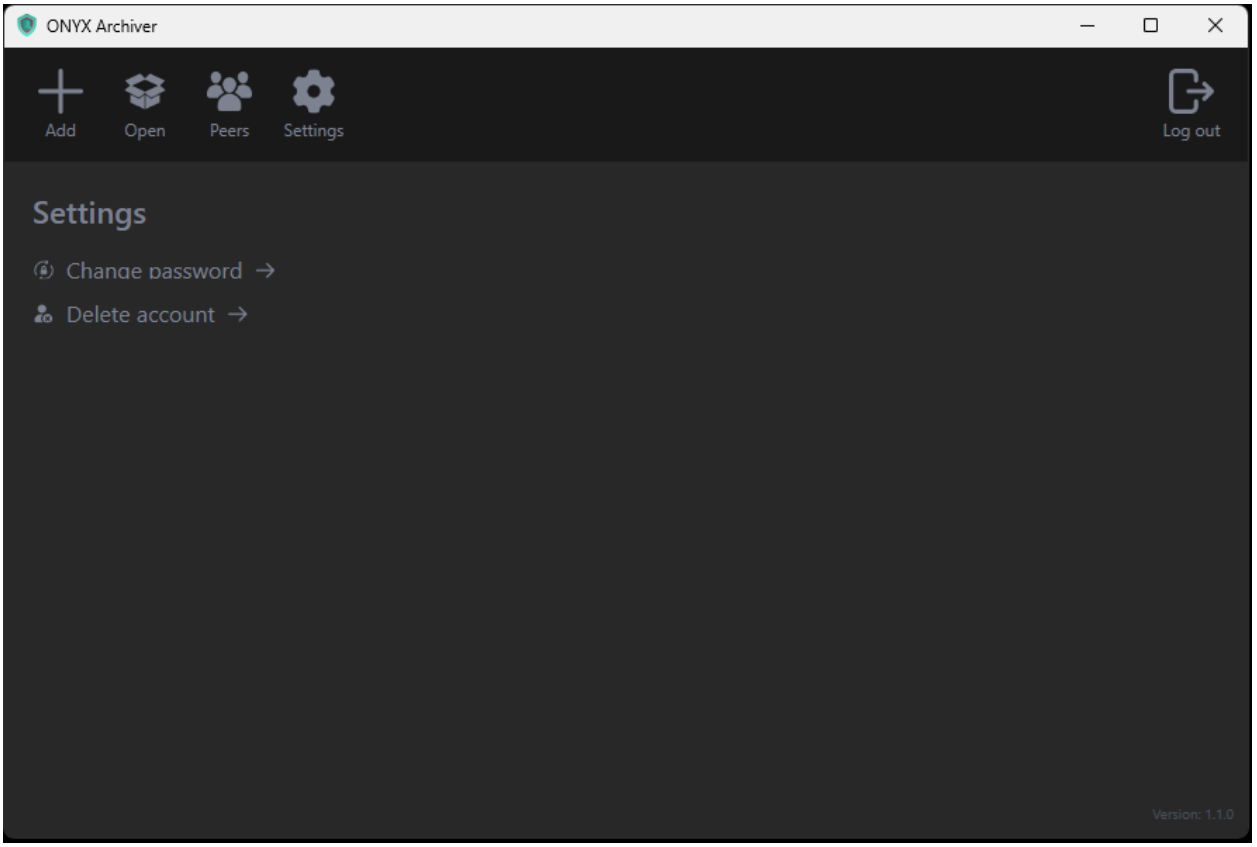

ONYX Archiver
Create account
User
User1111*
Create a new account
Authentication
Version: 1.1.0

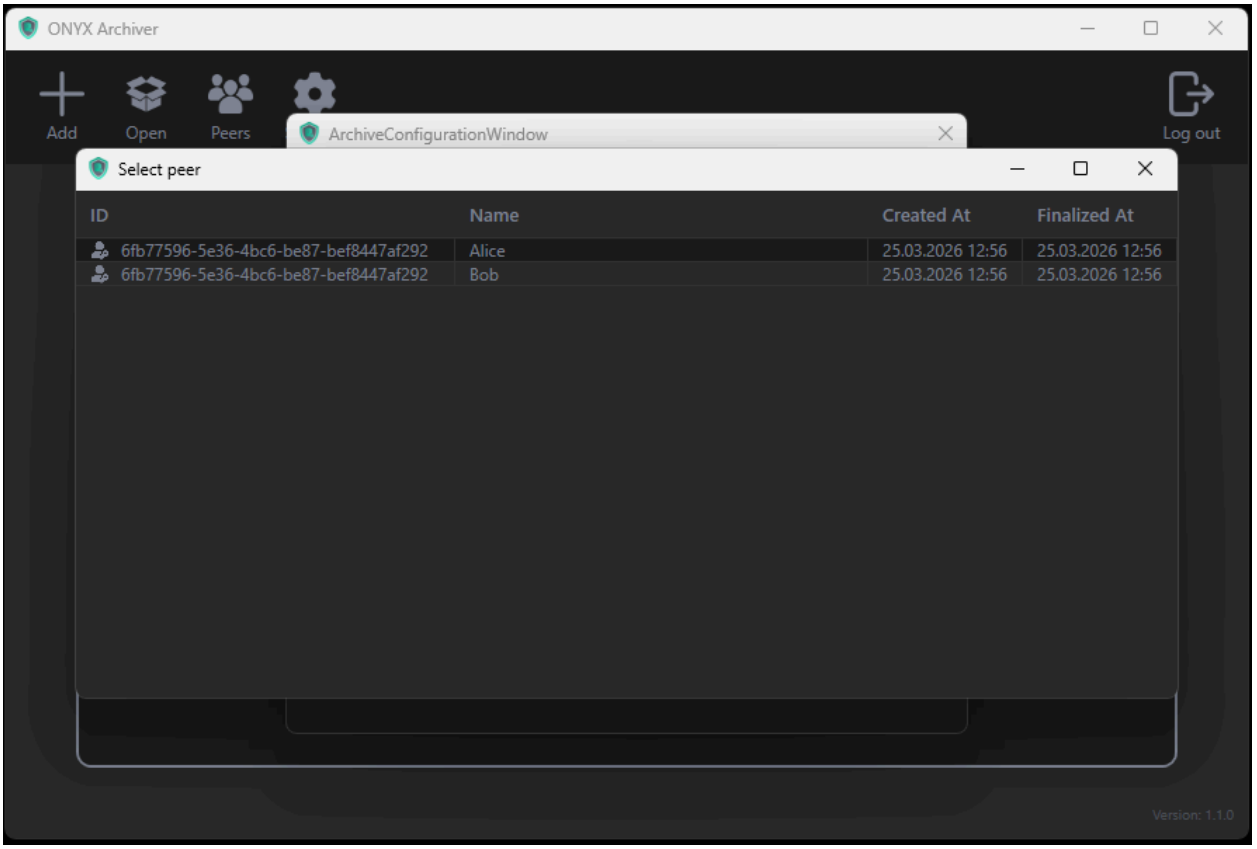

ONYX Archiver
Authentication
User
Log in
Create a new account
Version: 1.1.0

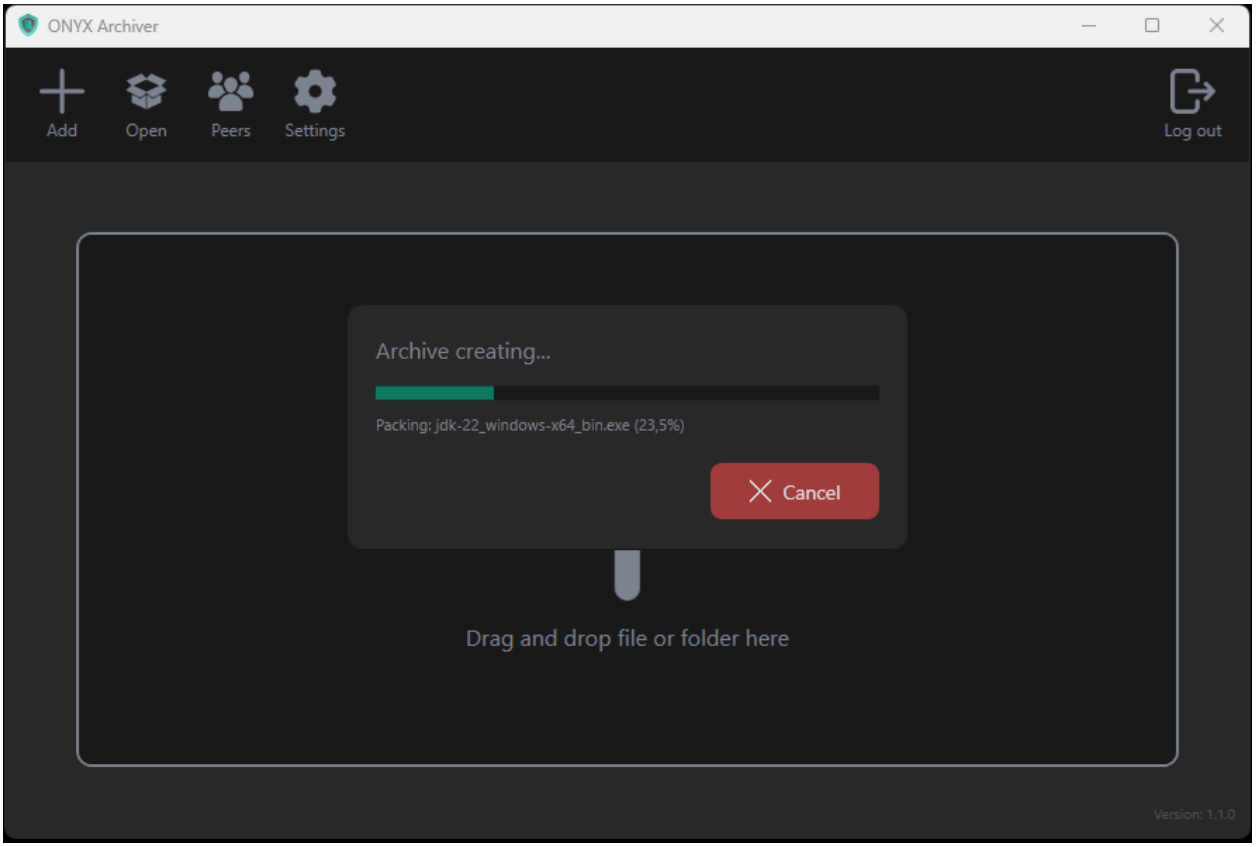

ONYX Archiver
Add
Open
Peers
Settings
Log out
Drag and drop file or folder here
Version: 1.1.0

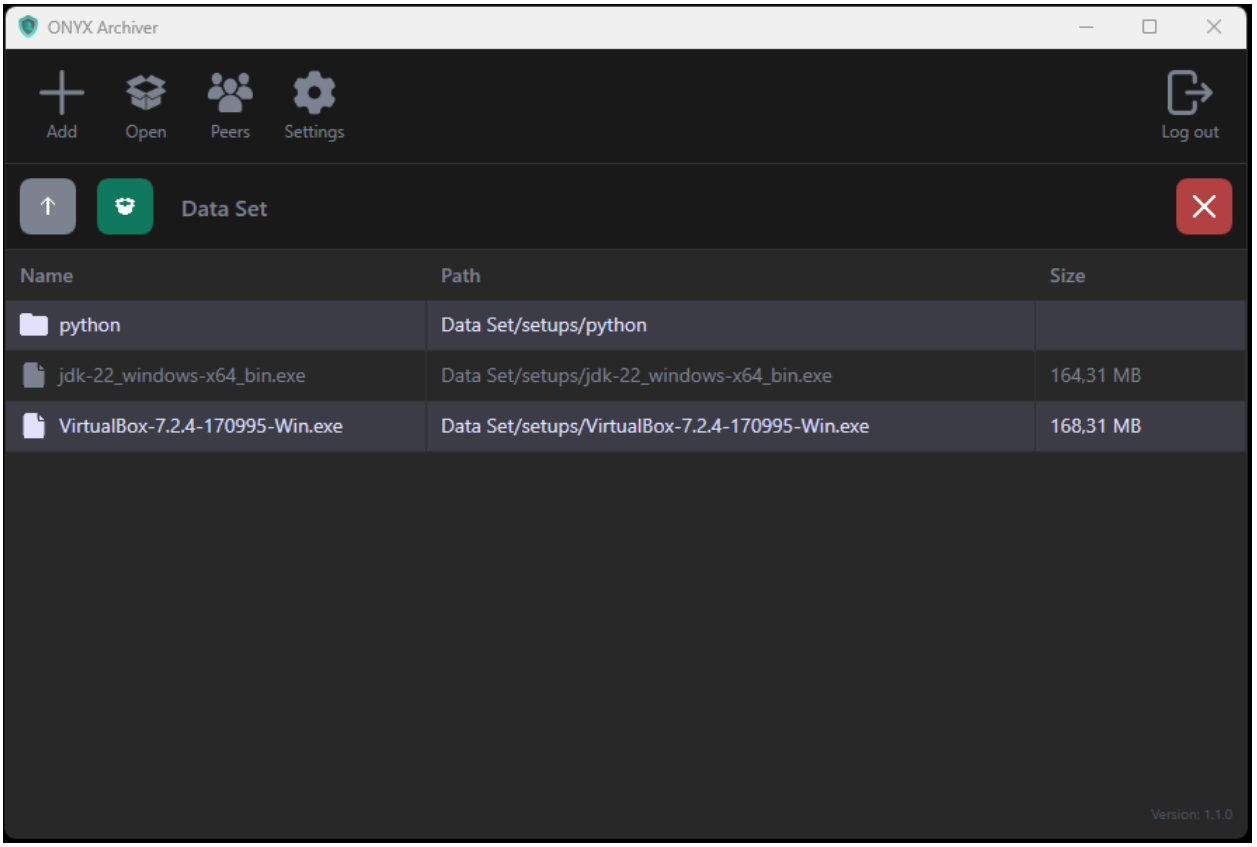

ONYX Archiver
Add
Open
Peers
Settings
Log out
Drag and drop archive file here
Version: 1.1.0

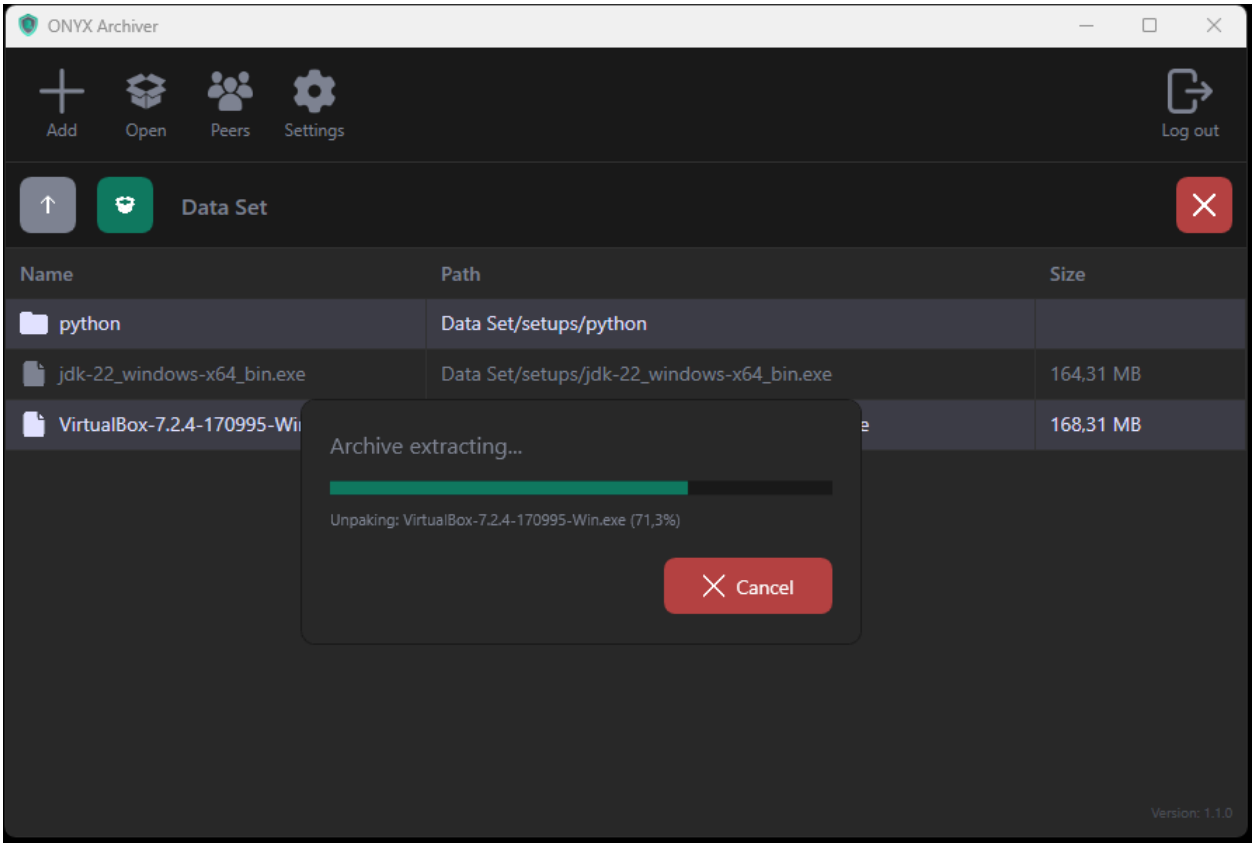

ONYX Archiver
Add
Open
Peers
Settings
Log out
ID
Name
Created At
Finalized At
Actions
6fb77596-5e36-4bc6-be87-bef8447af29
Alice
25.03.2026 12:56
25.03.2026 12:56
6fb77596-5e36-4bc6-be87-bef8447af29
Bob
25.03.2026 12:56
25.03.2026 12:56
Version: 1.1.0


ONYX Archiver
Add
Open
Peers
Settings
Log out
Settings
Change password
Delete account
Version: 1.1.0


ONYX Archiver
Add
Open
Peers
ArchiveConfigurationWindow
Log out
Select peer
ID
Name
Created At
Finalized At
6fb77596-5e36-4bc6-be87-bef8447af292
Alice
25.03.2026 12:56
25.03.2026 12:56
6fb77596-5e36-4bc6-be87-bef8447af292
Bob
25.03.2026 12:56
25.03.2026 12:56
Version: 1.1.0


ONYX Archiver
Add
Open
Peers
Settings
Log out
Archive creating...
Packing: jdk-22_windows-x64_bin.exe (23,5%)
Cancel
Drag and drop file or folder here
Version: 1.1.0





The complete list of screenshots is located in the “screenshots” folder.